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BACKUP Methodology

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Objective

The aim of this document is to create a comprehensive enterprise-grade database backup methodology using both Microsoft SQL Server and PostgreSQL.

Change History

Version	Date	Created by	Approved by
1.0	01-01-2026	DBA Team	Omar Chemes

Projects

gCABA

PROD environment

Intro

- Backup database policy
 - Database backup Type & Frequency

Project	Environment	db Name	bkp Type	Frequency	Job Name	Time	Duration Time
gCABA	PROD	[urbetrack]	FULL	Saturday - Tuesday - Thursday	GCABA-BACKUP-URBETRACK-FULL-DIFF	12:10:00 AM	
gCABA	PROD	[urbetrack]	DIFF	Monday - Wednesday - Friday	GCABA-BACKUP-URBETRACK-FULL-DIFF	12:10:00 AM	
gCABA	PROD	[OTHERS]	FULL	Thursday - Saturday	GCABA-BACKUP-RESTO-FULL-DIFF	Doesn't have	
gCABA	PROD	[OTHERS]	DIFF	Sunday - Monday - Tuesday - Wednesday - Friday	GCABA-BACKUP-RESTO-FULL-DIFF	Doesn't have	

gCABA	PROD	[urbetrack]	LOG	Every day	GCABA-BACKUP-URBETRACK-TLOG	All day (every 15 min)
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- Database backup retention

Project	Environment	db Name	bkp Type	Retention	Retention Description	Job Name	Driver	Path
gCABA	PROD	[urbetrack]	FULL	Not Apply		GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_urbetrac
gCABA	PROD	[OTHERS]	FULL	Not Apply		GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_full.bak
gCABA	PROD	[urbetrack]	FULL	Apply	There is a retention file of 4 weeks (28 days)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_urbetrac
gCABA	PROD	[OTHERS]	FULL	Apply	There is a retention of 2 weeks on week (14 days)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_full.bak
gCABA	PROD	[urbetrack]	DIFF	Not Apply		GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_urbetrac
gCABA	PROD	[OTHERS]	DIFF	Not Apply		GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_diff.bak
gCABA	PROD	[urbetrack]	DIFF	Apply	There is a retention file of 4 weeks (28 days)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_urbetrac
gCABA	PROD	[OTHERS]	DIFF	Apply	There is a retention of 1 week on week (1 week)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_diff.bak
gCABA	PROD	[urbetrack]	LOG	Apply	There is a retention of 2 days (2 days)	GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_urbetrac
gCABA	PROD	[urbetrack]	LOG	Apply	There is a retention of one week (1 week)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_urbetrac

 Important

DIFF backups GCABA-BACKUP-URBETRACK-FULL-DIFF --> **EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-DELETE-H-ALL'**

- Storage
 - gCABA - CLARO
- Monitoring & Notification

Backup jobs are monitored through the **PRTG monitoring platform**, which provides visibility into their operational status.

Additionally, a daily email report is sent to the DevSecOps team containing the backup results along with their corresponding verification (VERIFY) outcomes. Regarding disk space utilization, notifications are sent via email to the DevSecOps team, and dedicated PRTG sensors with predefined threshold values are configured to generate alerts whenever available disk space falls below acceptable levels.

- Security Management
 - Backup Encryption

Implement encryption for backup files to protect sensitive data and ensure the confidentiality of stored information.
 - Access Control

Ensure that only authorized personnel have access to backup files and related backup management processes.

Disabled PROD MSSQL Server Jobs

On this section we are going to describe the jobs that can be deleted from the instance.

- backup_report_tape
 - Status

Disabled
 - Last execution Date

12/25/2024 10:00:00 AM
 - Description

The job doesn't have a description.
 - Command

You can view the SQL script [here](#).
- BACKUP-PURGE
 - Status

Disabled
 - Last execution Date

3/5/2025 9:00:00 PM
 - Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

 1. Mount a K drive
 2. Get and Delete FULL backups files for URBETRACK Database older than 28 days
 3. Get and Delete DIFF backups files for URBETRACK Database older than 28 days
 4. Get and Delete LOG backups files for URBETRACK Database older than 7 days
 5. Get and Delete FULL backups files for others Databases older than 14 days
 6. Get and Delete DIFF backups files for others Databases older than 7 days
 7. Send eMail to geponce62@gmail.com
 8. Title PT-GCABA BACKUP-URBETRACK-PURGE
 - Command

You can view the SQL script [here](#).
- BACKUP-REPORTE-GCABA
 - Status

Disabled
 - Last execution Date

3/6/2025 8:15:00 AM
 - Description

The job doesn't have a description.

Reading the code we can mention that this Job execute a Report base on the .txt created on path **E:\REPORTES\BACKUP-REPORTE_BACKUPS_EN_DISCO.txt**
 - Command

- You can view the SQL script [here](#).
- BACKUP-RESTO-FULL-DIFF
 - Status
Disabled
 - Last execution Date
3/6/2025 3:46:12 AM
 - Description
The job doesn't have a description.
Reading the code we can mention that this Job execute a backup of others databases
 - Command
You can view the SQL script [here](#).
 - BACKUP-URBETRACK-FULL-DIFF
 - Status
Disabled
 - Last execution Date
3/6/2025 12:10:01 AM
 - Description
The job doesn't have a description.
Reading the code we can mention that this Job execute a FULL/DIFF backup of urbetrack with a specific schedule
`/* 1 Dom 2 Lun 3 Mar 4 Mie 5 Jue 6 Vie 7 Sab Base 7Sab 1Dom 2Lun 3Mar 4Mie 5Jue 6Vie urbetrack FULL DIFF DIFF FULL DIFF FULL DIFF */`
Finally move the files to H: FULL COPY_ONLY and execute the job BACKUP-RESTO-FULL-DIFF
 - Command
You can view the SQL script [here](#).
 - BACKUP-URBETRACK-TLOG
 - Status
Disabled
 - Last execution Date
5/27/2026 10:31:59 AM --> failed 3/6/2025 4:45:00 PM --> successfull
 - Description
The job doesn't have a description.
Reading the code we can mention that this Job execute a LOG backup of urbetrack.
 - Command
You can view the SQL script [here](#).

Enabled PROD environment

On this section we are going to describe the jobs that are running and taking the databases backups in PROD gCABA environment.

- GCABA_BACKUP_REPORTE_DISCOS_H_K
 - Status
Enabled
 - Last execution Date
6/1/2026 5:00:00 PM
 - Description
The job doesn't have a description.
Reading the code we can mention that this Job:
Step I:
Execute a PowerShell to REPORT the .bak files only (not .log files) from `H:\BACKUP_GCABA\bkp$fecha.bak**` and `K:\BACKUP_GCABA\bkp$fecha.bak**` drivers.

Step II:

Send email to infra@urbetrack.com; geponce62@gmail.com @file_attachments =
E:\REPORTES\GCABA_BACKUP_REPORTES\DISCOS_H_K_AT_1700.txt



Tip

This Job is only for reporting. Doesn't take any action

- Schedule

Occurs every day at 12:00:00 PM. Schedule will be used starting on 3/31/2025. Occurs every day at 5:00:00 PM. Schedule will be used starting on 3/31/2025.

- Command

You can view the SQL script [here](#).

- GCABA-BACKUP-COPY-TO-NFS

- Status

Enabled

- Last execution Date

3/28/2025 12:10:49 AM More than a year ago

- Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

Step I:

Execute a PowerShell to copy the files from **\$Origen = "H:\BACKUP_GCABA"** to **\$Destino = "K:\BACKUP_GCABA"**. To execute this copy we are using ROBOCOPY.

Step II:

Execute a PowerShell to rename the .log file that exists on "C:\scripts\copy_to_nfs_from_SQL.log". Finally this step create a new file base on copy_to_nfs_from_SQL.log

- Schedule

This job doesn't have an Schedule

- Command

You can view the SQL script [here](#)

- GCABA-BACKUP-DELETE-H-ALL

- Status

Enabled

- Last execution Date

6/2/2026 12:10:01 AM

- Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

Step I:

DELETE files from **H:\BACKUP_GCABA**

- PURGE URBETRACK FULL

There is not retention we delete the files

Delete FULL backups from "**H:\BACKUP_GCABA\bkp_urbetrack_full_*.bak**"

- PURGE URBETRACK DIFF

There is not retention we delete the files

Delete DIFF backups from "**H:\BACKUP_GCABA\bkp_urbetrack_diff_*.bak**"

- PURGE URBETRACK LOG

There is a retention of two days

Delete TLOG (OLDER than 2 days) from "**H:\BACKUP_GCABA\bkp_urbetrack_log_*.bak**"

- PURGE OTHERS FULL

There is not retention we delete the files

Delete FULL backups from "H:\BACKUP_GCABA\bkp_full.bak"

- PURGE OTHERS DIFF

There is not retention we delete the files

Delete DIFF backups from "H:\BACKUP_GCABA\bkp_diff.bak"

- Schedule

This job doesn't have an Schedule

- Command

You can view the SQL script [here](#)

- GCABA-BACKUP-DELETE-H-ALL-20251213

- Status

Enabled

- Last execution Date

12/14/2025 12:10:01 AM Last execution was six months ago

- Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

Step I:

This Job execute the same steps as GCABA-BACKUP-DELETE-H-ALL. But, has the first two steps disabled, these steps are related to the FULL and DIFF backups for [URBETRACK] database

- Schedule

This job doesn't have an Schedule

- Command

You can view the SQL script [here](#)

- GCABA-BACKUP-PURGE-K-ALL

- Status

Enabled

- Last execution Date

6/1/2026 9:00:00 PM

- Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

Step I:

DELETE files from K:\BACKUP_GCABA

- PURGE URBETRACK FULL

There is a retention file of 4 weeks (28 days)

Delete FULL backups from "K:\BACKUP_GCABA\bkp_urbetrack_full_*.bak"

- PURGE URBETRACK DIFF

There is a retention file of 4 weeks (28 days)

Delete DIFF backups from "K:\BACKUP_GCABA\bkp_urbetrack_diff_*.bak"

- PURGE URBETRACK LOG

There is a retention of one week

Delete TLOG (OLDER than 1 week) from "K:\BACKUP_GCABA\bkp_urbetrack_log_*.bak"

- PURGE OTHERS FULL

There is a retention of 2 weeks on week (14 days).

Delete FULL backups from "K:\BACKUP_GCABA\bkp_full.bak"

- PURGE OTHERS DIFF

There is a retention of 1 week on week (1 week).

Delete DIFF backups from "K:\BACKUP_GCABA\bkp_diff.bak"

Step II:

Send email to: geponce62@gmail.com with @file_attachments = 'E:\REPORTES\BACKUP_PURGE_K_ALL.txt'

- Schedule

Occurs every day at 9:00:00 PM. Schedule will be used starting on 8/2/2024.

- Command

You can view the SQL script [here](#)

- GCABA-BACKUP-REPORTE

- Status

Enabled

- Last execution Date

6/2/2026 8:15:00 AM

- Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

Step I:

Using [master] database. Check if there is a backup running.

Save info in .txt file on: First. List all files that start with "bkp" order by date [/O-D] descending and redirects the output to a file on [E:\REPORTES\BACKUP_REPORTE_BACKUPS_EN_DISCO.txt] Note [>] is use to overwrite exec xp_cmdshell "dir H:\BACKUP_GCABA\bkp* /O-D > E:\REPORTES\BACKUP_REPORTE_BACKUPS_EN_DISCO.txt"

Second. Lists all transaction log backup files order by date [/O-D] descending and adds the output to the end of the existing file [E:\REPORTES\BACKUP_REPORTE_BACKUPS_EN_DISCO.txt] Note [>>] is use to add exec xp_cmdshell "dir H:\BACKUP_GCABA\urbetrack_*.trn /O-D >> E:\REPORTES\BACKUP_REPORTE_BACKUPS_EN_DISCO.txt"

The Final Result is, after both commands run E:\REPORTES\BACKUP_REPORTE_BACKUPS_EN_DISCO.txt contains a List of all files beginning with bkp, followed by a list of all .trn files matching urbetrack_*.trn

Bulk insert into a table: truncate table GPL.dbo.BACKUP_REPORTE_BACKUPS_EN_DISCO

BULK INSERT GPL.dbo.BACKUP_REPORTE_BACKUPS_EN_DISCO FROM
"E:\REPORTES\BACKUP_REPORTE_BACKUPS_EN_DISCO.txt";

Send email to infra@urbetrack.com; geponce62@gmail.com with the result of the queries:

```
/* URBETRACK db / select * from GPL.dbo.uv_BACKUP_REPORTE_BACKUPS_EN_DISCO_urbetrack order by 1' / REST of the dbs */  
select * from GPL.dbo.uv_BACKUP_REPORTE_BACKUPS_EN_DISCO_resto order by 1
```

- Schedule

Occurs every day at 8:15:00 AM. Schedule will be used starting on 3/7/2020.

- Command

You can view the SQL script [here](#)

- GCABA-BACKUP-RESTO-FULL-DIFF

- Status

Enabled

- Last execution Date

6/2/2026 12:34:14 AM

- Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

Step I:

Declare many variables Set @DEBUG = N, @CIFRADO = N, @FORZAR_TIPO_BACKUP = N, @TIPO_BACKUP_FORZADO = DIFF, @DISCO = H, @DIRECTORIO = \BACKUP_CABA\

If @cifrado is 'S' we are going to set up: set @subtexto='cifr' set @backup_with_full='WITH FORMAT, COMPRESSION, CHECKSUM, STATS=1, ENCRYPTION(ALGORITHM=AES_256, SERVER CERTIFICATE=CertificateForBackupPVK)' set @backup_with_diff='WITH FORMAT, COMPRESSION, CHECKSUM, STATS=1, DIFFERENTIAL, ENCRYPTION(ALGORITHM=AES_256, SERVER CERTIFICATE=CertificateForBackupPVK)' If not set @subtexto='_' set @backup_with_full='WITH FORMAT, COMPRESSION, CHECKSUM, STATS=1' set @backup_with_diff='WITH FORMAT, COMPRESSION, CHECKSUM, STATS=1, DIFFERENTIAL'

Set week day to take FULL or DIFF depend on the new schedule. If it's 1Dom 2Lun 4Mie 6Vie we will take FULL if not we will take DIFF. /--if @weekday in (1,2,4,6) -- FULL: 1Dom 2Lun 4Mie 6Vie/ if @weekday in (5,7) -- FULL: 5Jue 7Sab set @tipo_backup='FULL' /*--else -- DIFF: 3Mar 5Jue 7Sab */ else -- DIFF (1,2,3,4,6) 1Dom 2Lun 3Mar 4Mie 6Vie set @tipo_backup='DIFF'

if @forzar_tipo_backup='S' set @tipo_backup=@tipo_backup_forzado

If a backups is running we wait.

If @tipo_backup='FULL' We take FULL backup for the rest of the databases (not TEMPDB - URBETRACK). We create 3 .bak files per database We verified the .bak ELSE We take DIFF backup for the rest of the databases (not MASTER - TEMPDB - URBETRACK). We verified the .bak

The FULL and DIFF backups file are done in H:\BACKUP_GCABA

Step II:

Send email to geponce62@gmail.com in case of Succeed or Fail with @file_attachments = 'E:\REPORTES\GCABA-BACKUP-RESTO-FULL-DIFF.txt'

- Schedule

This job doesn't have an Schedule

- Command

You can view the SQL script [here](#)

- GCABA-BACKUP-URBETRACK-FULL-DIFF

- Status

Enabled

- Last execution Date

6/2/2026 12:10:00 AM

- Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

Step I:

Execute command on [master] database: **EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-DELETE-H-ALL'** Delete everything. The only one with retention is the [URBETRACK] LOG

Step II:

Declare many variables Set @DEBUG = N, @CIFRADO = N, @FORZAR_TIPO_BACKUP = N, @TIPO_BACKUP_FORZADO = DIFF, @DISCO = H, @DIRECTORIO = \BACKUP_CABA\

If @cifrado is 'S' we are going to set up: set @subtexto='cif' set @backup_with_full='WITH FORMAT, COMPRESSION, CHECKSUM, STATS=1, ENCRYPTION(ALGORITHM=AES_256, SERVER CERTIFICATE=CertificateForBackupPVK)' set @backup_with_diff='WITH FORMAT, COMPRESSION, CHECKSUM, STATS=1, DIFFERENTIAL, ENCRYPTION(ALGORITHM=AES_256, SERVER CERTIFICATE=CertificateForBackupPVK)' If not set @subtexto='_' set @backup_with_full='WITH FORMAT, COMPRESSION, CHECKSUM, STATS=1' set @backup_with_diff='WITH FORMAT, COMPRESSION, CHECKSUM, STATS=1, DIFFERENTIAL'

Set week day to take FULL or DIFF depend on the new schedule. If it's 7Sab 3Mar 5Jue we will take FULL, if not we will take DIFF. if @weekday in (7,3,5) -- FULL: Sab Mar Jue (7,3,5) set @tipo_backup='FULL' else -- DIFF: Dom Lun Mie Vie (1,2,4,6) set @tipo_backup='DIFF'

if @forzar_tipo_backup='S' set @tipo_backup=@tipo_backup_forzado

if @forzar_tipo_backup='S' set @tipo_backup=@tipo_backup_forzado

If a backups is running we wait.

If @tipo_backup='FULL' We take FULL backup for the [URBETRACK]. We create the .bak with up to 6 files. We verified the .bak ELSE We take DIFF backup for [URBETRACK] databases with up to 3 files. We verified the .bak

The FULL and DIFF backups file are done in H:\BACKUP_GCABA

Reattempt If we couldn't take the [URBETRACK] database backup we reattempt until reach the limit. The limit is @HHMM_limite='05:00'

Step III & IV:

Send email to geponce62@gmail.com in case of Succeed or Fail with @file_attachments = 'E:\REPORTES\GCABA-BACKUP-RESTO-FULL-DIFF.txt'

Step V:

Execute command on [master] database: **EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-RESTO-FULL-DIFF'** Take FULL and DIFF backup of the rest of the databsaes.

Step VI:

```
-- DESHABILITADO -- Acordado en la reunion con Nico, Ramiro, David del Viernes 28/03/2025 -- La copia de los backups del
H: al NFS se hace con ROBOCOPY por afuera de SQL Server -- El script de copia con ROBOCOPY lo programa Nico en otro
scheduler -- EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-COPY-TO-NFS' print 'Este Job de copia al NFS queda
DESHABILITADO => EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-COPY-TO-NFS' print 'Acordado en la reunion con Nico,
Ramiro, David del Viernes 28/03/2025'
```

Step VI:

```
-- 2025-0416 -- Dejo desactivado el purgado de los backups del NFS -- Nico se encarga de la depuracion de los backups del
NS en GCABA
```

```
-- EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-PURGE-K-ALL'
```

```
-- EVIDENCIA -- ===== -- GCABA/CLARO - Purgado de los backups de los LOG de la base urbetrack del NFS
```

```
-- Guillermo Ponce de Leon geponce62@gmail.com 14 de abril de 2025, 19:43 -- Para: Nicolás Zabaleta
nicolas.zabaleta@urbetrack.com -- Hola Nico. -- De acuerdo a lo conversado por whatsapp deshabilité en GCABA/CLARO el
purgado de los backup de los LOG de la base urbetrack en el NFS. -- Saludos. -- ===== -- Nicolás
Zabaleta nicolas.zabaleta@urbetrack.com 15 de abril de 2025, 10:00 -- Para: Guillermo Ponce de Leon geponce62@gmail.com
-- Buen dia Guille, como va? -- Perfecto gracias, hay algún otro job que utilice el NFS? porque debido al ultimo cambio de la
forma de copy, estamos analizando dejar el nfs en solo lectura, ya -- que ese no se utiliza para la web. -- Avisame cualquier
cosa, saludos. -- ===== -- Guillermo Ponce de Leon geponce62@gmail.com 15 de abril de 2025,
19:14 -- Para: Nicolás Zabaleta nicolas.zabaleta@urbetrack.com -- Hola Nico. -- El Job que depura los backups viejos (FULL y
DIFF) accede al NFS. -- Saludos. -- ===== -- Nicolás Zabaleta nicolas.zabaleta@urbetrack.com 16
de abril de 2025, 9:57 -- Para: Guillermo Ponce de Leon geponce62@gmail.com -- Buen dia Guille, como estas? -- Si estas
hablando de claro/gcaba entonces ese job tambien hay que desactivarlo porque la retencion la estamos manejando
directamente desde el NFS de la misma -- -- forma que los logs. -- Saludos
```

- Schedule

Occurs every day at 12:10:00 AM. Schedule will be used starting on 12/15/2025.*

- Command

You can view the SQL script [here](#)

- GCABA-BACKUP-URBETRACK-FULL-DIFF-20251213

- Status

Enabled

- Last execution Date

12/14/2025 12:10:00 AM Last execution six months ago

- Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

Step I:

Execute command on [master] database: **EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-DELETE-H-ALL-20251213'**

Step II:

Set select @forzar_tipo_backup="N", Set @tipo_backup_forzado="DIFF" Set select @disco="H:",
@directorio="\\BACKUP_GCABA" -- H:\\BACKUP_GCABA Set var @cifrado for FULL and DIFF backup. If @cifrado is 'S' we are
going to use ENCRYPTION(ALGORITHM=AES_256, SERVER CERTIFICATE=CertificateForBackupPVK) Set @weekday Set
@tipo_backup Check @debug Take FULL backup of [URBETRACK] database in 6 files and verify it Take DIFF backup of
[URBETRACK] database in 3 files and verify it

REINTENTAR_BACKUP: **The limit of the attempts is the TIME LIMIT (@reintentos).**

Step III/IV:

Send email in case of Succes or Fail to geponce62@gmail.com

Step V:

Execute command on [master] database: **EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-RESTO-FULL-DIFF'**

Step VI:

```
-- DESHABILITADO -- Acordado en la reunion con Nico, Ramiro, David del Viernes 28/03/2025 -- La copia de los backups del
H: al NFS se hace con ROBOCOPY por afuera de SQL Server -- El script de copia con ROBOCOPY lo programa Nico en otro
scheduler -- EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-COPY-TO-NFS' print 'Este Job de copia al NFS queda
DESHABILITADO => EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-COPY-TO-NFS' print 'Acordado en la reunion con Nico,
Ramiro, David del Viernes 28/03/2025'
```

Step VII:

-- 2025-0416 -- Dejo desactivado el purgado de los backups del NFS -- Nico se encarga de la depuracion de los backups del NS en GCABA

-- EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-PURGE-K-ALL'

-- EVIDENCIA -- ===== -- GCABA/CLARO - Purgado de los backups de los LOG de la base urbetrack del NFS

-- Guillermo Ponce de Leon geponce62@gmail.com 14 de abril de 2025, 19:43 -- Para: Nicolás Zabaleta nicolas.zabaleta@urbetrack.com -- Hola Nico. -- De acuerdo a lo conversado por whatsapp deshabilité en GCABA/CLARO el purgado de los backup de los LOG de la base urbetrack en el NFS. -- Saludos. -- ===== -- Nicolás Zabaleta nicolas.zabaleta@urbetrack.com 15 de abril de 2025, 10:00 -- Para: Guillermo Ponce de Leon geponce62@gmail.com -- Buen dia Guille, como va? -- Perfecto gracias, hay algún otro job que utilice el NFS? porque debido al ultimo cambio de la forma de copy, estamos analizando dejar el nfs en solo lectura, ya -- que ese no se utiliza para la web. -- Avisame cualquier cosa, saludos. -- ===== -- Guillermo Ponce de Leon geponce62@gmail.com 15 de abril de 2025, 19:14 -- Para: Nicolás Zabaleta nicolas.zabaleta@urbetrack.com -- Hola Nico. -- El Job que depura los backups viejos (FULL y DIFF) accede al NFS. -- Saludos. -- ===== -- Nicolás Zabaleta nicolas.zabaleta@urbetrack.com 16 de abril de 2025, 9:57 -- Para: Guillermo Ponce de Leon geponce62@gmail.com -- Buen dia Guille, como estas? -- Si estas hablando de claro/gcaba entonces ese job tambien hay que desactivarlo porque la retencion la estamos manejando directamente desde el NFS de la misma -- -- forma que los logs. -- Saludos */

- o Schedule

Occurs every day at 12:10:00 AM. Schedule will be used starting on 12/15/2025.*

- o Command

You can view the SQL script [here](#)

- GCABA-BACKUP-URBETRACK-TLOG

- o Status

Enabled

- o Last execution Date

6/2/2026 3:45:00 PM

- o Description

The job doesn't have a description.

Reading the code we can mention that this Job execute:

Step I:

Set dates variables Execute [URBETRACK] LOG database backup on

H:\BACKUP_GCABA\bkp_urbetrack_log_"+@YYYYMMDD+"_"+@HHMMSS+".bak

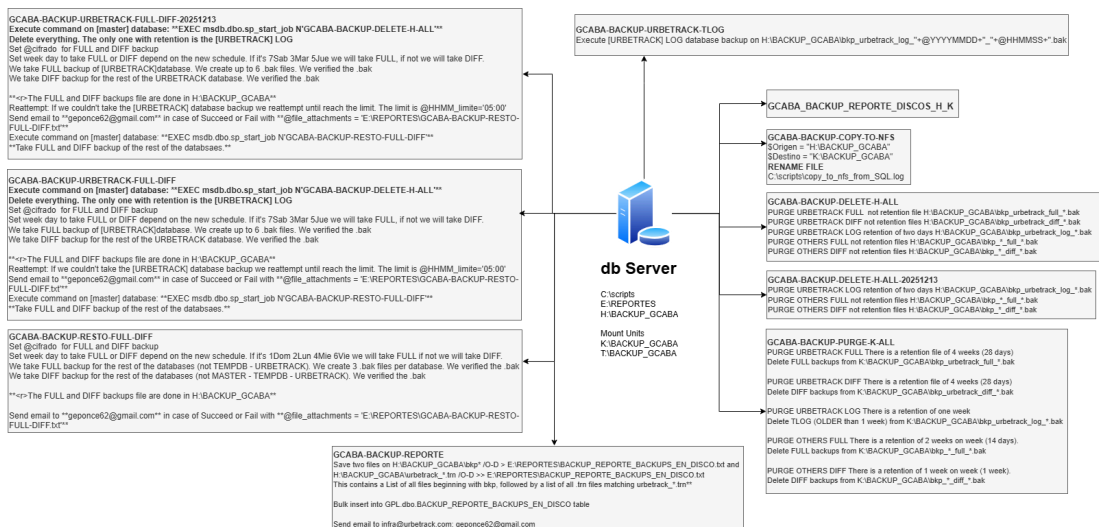
- o Schedule

Occurs every day every 15 minute(s) between 12:00:00 AM and 11:59:59 PM. Schedule will be used starting on 8/4/2024.*

- o Command

You can view the SQL script [here GCABA-BACKUP-URBETRACK-TLOG.sql](#)

- Diagram



Questions & Meetings

On this section we are going to give a detail of the question that we have after a tudy done of backups in gCABA for PROD environment.

GUILLE

- Can we delete the six JOBS that are disable on PROD gCABA? ([backup_report_tape] - [BACKUP-PURGE] - [BACKUP-REPORTE-GCABA] - [BACKUP-RESTO-FULL-DIFF] - [BACKUP-URBETRACK-FULL-DIFF] - [BACKUP-URBETRACK-TLOG]) BACKUPAR Y ELIMINAR DE LA INSTANCIA!!!!!!
- [GCABA_BACKUP_REPORTE_DISCOS_H_K]
 - Who wrote the files on *H:\BACKUP_GCABA\bkp\$fecha.bak*** and *K:\BACKUP_GCABA\bkp\$fecha.bak*** and **'E:\REPORTES\GCABA_BACKUP_REPORTE_DISCOS_H_K_AT_1700.txt'**?
 - Are we deleteing or steping on the files on **E:\REPORTES**?

En el H solo tenemos espacio para los backups del día El historico esta en el K

- [GCABA-BACKUP-COPY-TO-NFS]
 - This job was executed for last time more than a year ago 3/28/2025 12:10:49 AM. Can we delete it?
 - This job was executed for last time more than a year ago, but if I connect to "C:\scripts" I can see the file "copy_to_nfs_from_SQL.log" created today (6/3/2026). So, who created this file? and for what?
- [GCABA-BACKUP-DELETE-H-ALL]
 - I don't have any question. We delete everything less LOG that we keep the last two days
- [GCABA-BACKUP-DELETE-H-ALL-20251213]
 - Last execution was six months ago 12/14/2025 12:10:01 AM. Can we delete it? backpear y eliminar
- [GCABA-BACKUP-PURGE-K-ALL]
 - Why we have a retention file here and not in the H drive?
- [GCABA-BACKUP-REPORTE]
 - No questions
- [GCABA-BACKUP-RESTO-FULL-DIFF]
 - We don't have a schedule for this, why? - Because is executed by [GCABA-BACKUP-URBETRACK-FULL-DIFF] the next job on the list
 - Why we take 4 FULL backups per week?
 - Why we create more than a file per database that is not [TEMPDB] - [URBETRACK]. Databases are really small.
- [GCABA-BACKUP-URBETRACK-FULL-DIFF]
 - Same point before. Why we create up to 6/3 files manually?
- [GCABA-BACKUP-URBETRACK-FULL-DIFF-20251213]
 - Last execution was six months ago 12/14/2025 12:10:00 AM. Can we delete it?

backpear y eliminar

- [GCABA-BACKUP-URBETRACK-TLOG]
 - No questions
 - **46379 TICKET ZOHO VER QUILOMBO DEL DIA 12/14/2025 - MODIFICACION DE TIPO DE DATO INT TO BIGINT**
- TELEFONICA HACE 6 MESES NO SE HACEN CORRECTAMENTE LOS BACKUPS.
- Nico dar permisos a CloudFlare

DAVID

- [[https://dev.azure.com/urbetrack/SRE Team/_wiki/wikis/SRE-Team.wiki/2325/Procedimiento-de-Copias-de-Seguridad-y-Restauracion](https://dev.azure.com/urbetrack/SRE%20Team/_wiki/wikis/SRE-Team.wiki/2325/Procedimiento-de-Copias-de-Seguridad-y-Restauracion)]
 - From this document the section [storage] for [CLARO] is talking about LINUX but it's a WINDOWS.
 - [PRTG] - Request permissions to PRTG to Rodrigo?? JUMPSERVER CON 365 --> se saco y se puso atras de cloudflare --> se rompio Se sedio el trafico a cloudflare porque es gratuita Migrar a Prometheus y Grafana, esto era parte del laburo de Nico. HABLAR CON OMAR ESY NICO ESTE TEMA!!
 - [BaaS Telefónica VDC] - It's only for TELEFONICA not for CLARO

TELEFONICA

PROD environment

Intro

- Backup database policy

- Database backup Type & Frequency

Project	Environment	db Name	bkp Type	Frequency	Job Name	Time	Duration Time
gCABA	PROD	[urbetrack]	FULL	Saturday - Tuesday - Thursday	GCABA-BACKUP-URBETRACK-FULL-DIFF	12:10:00 AM	
gCABA	PROD	[urbetrack]	DIFF	Monday - Wednesday - Friday	GCABA-BACKUP-URBETRACK-FULL-DIFF	12:10:00 AM	
gCABA	PROD	[OTHERS]	FULL	Thursday - Saturday	GCABA-BACKUP-RESTO-FULL-DIFF	Doesn't have	
gCABA	PROD	[OTHERS]	DIFF	Sunday - Monday - Tuesday - Wednesday - Friday	GCABA-BACKUP-RESTO-FULL-DIFF	Doesn't have	
gCABA	PROD	[urbetrack]	LOG	Every day	GCABA-BACKUP-URBETRACK-TLOG	All day (every 15 min)	

- Database backup retention

Project	Environment	db Name	bkp Type	Retention	Retention Description	Job Name	Driver	Path
gCABA	PROD	[urbetrack]	FULL	Not Apply		GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_urbetrack
gCABA	PROD	[OTHERS]	FULL	Not Apply		GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_full.bak
gCABA	PROD	[urbetrack]	FULL	Apply	There is a retention file of 4 weeks (28 days)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_urbetrack
gCABA	PROD	[OTHERS]	FULL	Apply	There is a retention of 2 weeks on week (14 days)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_full.bak
gCABA	PROD	[urbetrack]	DIFF	Not Apply		GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_urbetrack
gCABA	PROD	[OTHERS]	DIFF	Not Apply		GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_diff.bak
gCABA	PROD	[urbetrack]	DIFF	Apply	There is a retention file of 4 weeks (28 days)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_urbetrack
gCABA	PROD	[OTHERS]	DIFF	Apply	There is a retention of 1 week on week (1 week)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_diff.bak

gCABA	PROD	[urbetrack]	LOG	Apply	There is a retention of 2 days (2 days)	GCABA-BACKUP-DELETE-H-ALL	H	H:\BACKUP_GCABA\bkp_urbetrac
gCABA	PROD	[urbetrack]	LOG	Apply	There is a retention of one week (1 week)	GCABA-BACKUP-PURGE-K-ALL	K	K:\BACKUP_GCABA\bkp_urbetrac

Important

DIFF backups GCABA-BACKUP-URBETRACK-FULL-DIFF --> **EXEC msdb.dbo.sp_start_job N'GCABA-BACKUP-DELETE-H-ALL'**

- Storage

- gCABA - CLARO

- Monitoring & Notification

Backup jobs are monitored through the **PRTG monitoring platform**, which provides visibility into their operational status.

Additionally, a daily email report is sent to the DevSecOps team containing the backup results along with their corresponding verification (VERIFY) outcomes. Regarding disk space utilization, notifications are sent via email to the DevSecOps team, and dedicated PRTG sensors with predefined threshold values are configured to generate alerts whenever available disk space falls below acceptable levels.

- Security Management

- Backup Encryption

Implement encryption for backup files to protect sensitive data and ensure the confidentiality of stored information.

- Access Control

Ensure that only authorized personnel have access to backup files and related backup management processes.

Disabled PROD MSSQL Server Jobs

On this section we are going to describe the jobs that can be deleted from the instance.

- **_OLD BACKUP-FULL-EXCEPTO-URBETRACK-URBE_HISTORY**
 - Status **Disabled**
 - Last execution Date **8/23/2024 4:15:00 AM**
 - Description **The job doesn't have a description.**
 - Command You can view the SQL script [here](./TELEFONICA/_OLD BACKUP-FULL-EXCEPTO-URBETRACK-URBE_HISTORY.sql).
- **_OLD BACKUP-FULL-EXCEPTO-URBETRACK-URBE_HISTORY-VERIFY**
 - Status **Disabled**
 - Last execution Date **8/23/2024 10:00:00 AM**
 - Description **The job doesn't have a description.**
 - Command You can view the SQL script [here](./TELEFONICA/_OLD BACKUP-FULL-EXCEPTO-URBETRACK-URBE_HISTORY-VERIFY.sql).
- **_OLD BACKUP-URBE_HISTORY-DIFF**
 - Status **Disabled**
 - Last execution Date **8/23/2024 2:15:00 AM**
 - Description **The job doesn't have a description.**
 - Command You can view the SQL script [here](./TELEFONICA/_OLD BACKUP-URBE_HISTORY-DIFF.sql).
- **_OLD BACKUP-URBE_HISTORY-FULL**
 - Status **Disabled**
 - Last execution Date **8/17/2024 11:45:00 PM**
 - Description **The job doesn't have a description.**
 - Command You can view the SQL script [here](./TELEFONICA/_OLD BACKUP-URBE_HISTORY-FULL.sql).
- **_OLD BACKUP-URBETRACK-DIFF**

- Status **Disabled**
- Last execution Date **8/17/2024 11:45:00 PM**
- Description **The job doesn't have a description.**
- Command You can view the SQL script [\[here\]\(./TELEFONICA/_OLD BACKUP-URBETRACK-DIFF.sql\)](#).
- **_OLD BACKUP-URBETRACK-FULL**
 - Status **Disabled**
 - Last execution Date **8/17/2024 2:00:00 PM**
 - Description **The job doesn't have a description.**
 - Command You can view the SQL script [\[here\]\(./TELEFONICA/_OLD BACKUP-URBETRACK-FULL.sql\)](#).
- **_OLD BACKUP-URBETRACK-LOG**
 - Status **Disabled**
 - Last execution Date **8/11/2024 7:30:00 PM**
 - Description **The job doesn't have a description.**
 - Command You can view the SQL script [\[here\]\(./TELEFONICA/_OLD BACKUP-URBETRACK-LOG.sql\)](#).

Enabled PROD environment

On this section we are going to describe the jobs that are running and taking the databases backups in PROD gCABA environment.

- **_BACKUP-RESTO-FULL-DIFF-OneTime**
 - Status **Enabled**
 - Last execution Date **6/30/2025 5:36:23 PM**
 - Description **The job doesn't have a description.**
 - Command You can view the SQL script [here](#).
- **BACKUP_LISTADO_DIARIO**
 - Status **Enabled**
 - Last execution Date **6/5/2026 3:00:00 AM**
 - Description **The job doesn't have a description.**

Reading the code we can mention that this Job execute:

Step I:

 - Command You can view the SQL script [here](#).

Questions & Meetings

- Can we backup and delete all the Disabled jobs?
- **[_BACKUP-RESTO-FULL-DIFF-OneTime]**
 - Last execution time was almost a year ago. Can we backup and delete it?

Hablar con Omar por el error en los jobs de telefonica. Guille ya te paso los certificados y quedo en la nada.

diferencia del TRLog log shipping

1. Arreglar los JOBS de verified (Guille mandame mail) + Hablarlo con Omar el Lunes
2. PRTG --> Solucionar lo que no esta funcionando
3. BKP y eliminar todo lo deshabilitado para gCABA y RED
4. prometheus + grafana
5. Implementar Ola Hallegren

Omar esto no funciona dba@urbetrack.com no crea Ticket

Proposal

- Summary

This assessment reviewed the current SQL Server backup implementation used for the gCABA environment and identified a mature but heavily customized backup framework based on SQL Agent Jobs, T-SQL scripts, PowerShell automation, file-system operations, NFS storage, reporting jobs, and restore validation processes.

The current solution demonstrates several positive operational practices, including:

- Full, Differential, and Transaction Log backups
- Automated backup cleanup
- Restore validation jobs
- Backup reporting
- Secondary storage using NFS repositories

However, the environment also presents significant opportunities for improvement:

- Extensive use of custom code and xp_cmdshell
- Lack of backup encryption
- Limited centralized monitoring
- Lack of SLA-based alerting
- Dependence on custom retention scripts
- Limited observability capabilities
- Potential recovery and compliance risks

This document proposes a modern enterprise backup architecture based on:

- SQL Server 2022 Best Practices (future implementation)
- Ola Hallengren Maintenance Solution
- Prometheus Monitoring
- Grafana Visualization
- AlertManager Alerting
- Backup Encryption
- Automated Recovery Validation
- Disaster Recovery Readiness

1. Current State Assessment

- Current Architecture Overview

Current backup architecture consists of:

SQL Server
|

SQL Agent Jobs | Full/Diff Backups Log Backups Restore Validation Cleanup Jobs Reporting Jobs | Local Backup Disk (H:) | NFS Copy (K:) | Long-Term Storage

- Backup Types Identified

Full Backups

Purpose: - Complete database backup - Foundation for recovery chain

Observed: - Managed through custom jobs

Differential Backups Purpose: - Reduce restore time - Reduce backup size

Observed: - Combined with Full backup framework

Transaction Log Backups Purpose: - Point-in-time recovery

Observed: - Dedicated SQL Agent jobs

Restore Validation Purpose: - Verify backup usability

Observed: - Dedicated restore testing jobs

- Storage Locations

Primary Storage --> H:\BACKUP_GCABA Secondary Storage --> K:\BACKUP_GCABA Mounted from: \ARO-NFS\backup\dbbackups Reporting --> E:\REPORTES

- Strengths

Good Backup Coverage (Full Differential Log) Restore Validation (One of the most valuable practices already implemented)

Secondary Repository (NFS copy process provides redundancy) Reporting (Backup reporting already exists) Retention Management (Automated cleanup processes implemented)

- Risks and Weaknesses

Single Points of Failure Local Backup Storage H:\BACKUP_GCABA Risk: Disk failure, Corruption, Capacity exhaustion

SQL Agent Dependency All operations depend on SQL Agent. Risk: Agent outage stops backups

Security Risks, xp_cmdshell Usage Current jobs execute operating system commands. Risks: Privilege escalation, Compliance findings, Attack surface expansion Recommendation: Disable after migration.

No Encryption Current backups appear unencrypted. Risk: Data exposure, Regulatory violations, Monitoring Risks

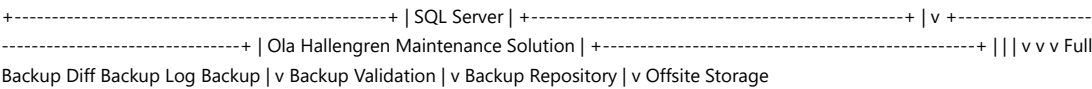
No centralized monitoring Potential issues remain undetected: Missed backups, Storage shortages, Long-running jobs

2. Gap Analysis

Area	Current State	Recommended State
Backup Framework	Custom Scripts	Ola Hallengren
Encryption	Not Implemented	AES-256
Monitoring	Reports Only	Prometheus
Visualization	Text Reports	Grafana
Alerting	Basic	AlertManager
Validation	Partial	Automated Restore Testing
Compliance	Limited	Auditable
Observability	Minimal	Enterprise Grade
Retention	Custom Scripts	Policy Driven
Recovery Metrics	Not Measured	RPO/RTO Monitoring

3. Proposed Backup Architecture

- Core Components



- Backup Strategy

FULL Backups

Setting	Value
Frequency	Weekly
Schedule	Sunday 01:00
Compression	Enabled
Encryption	Enabled
Verification	Enabled

```
EXECUTE dbo.DatabaseBackup
    @Databases = 'USER_DATABASES',
    @Directory = '\\BackupRepository\SQL',
    @BackupType = 'FULL',
    @Compress = 'Y',
    @Verify = 'Y';
```

Differential Backups

Setting	Value
Frequency	Daily
Schedule	01:00
Compression	Enabled

Transaction Log Backups

Setting	Value
Frequency	Every 15 Minutes
Recovery Model	FULL
Verification	Enabled

System Database Backups Daily: master, msdb, model

Recommended Schedule: Daily 02:00

- Retention Policy

Daily Backups Retention: 35 Days Reason: Supports operational recovery.

Weekly Backups Retention: 12 Weeks Reason: Supports historical recovery.

Monthly Backups Retention: 13 Months Reason: Annual auditing requirements.

Yearly Backups Retention: 7 Years Reason: Compliance and legal requirements.

- Recovery Strategy

Recovery Objectives

Metric	Target
RPO	15 Minutes
RTO	2 Hours

Recovery Scenarios Database Restore

Steps: Restore Full Backup Restore Differential Restore Logs Recover Database

- Security Recommendations

Backup Encryption

Create certificate:

```
CREATE MASTER KEY ENCRYPTION BY PASSWORD='StrongPassword';

CREATE CERTIFICATE BackupCertificate
WITH SUBJECT='Backup Encryption';
```

Backup:

```
WITH COMPRESSION,
ENCRYPTION
(
    ALGORITHM = AES_256,
    SERVER CERTIFICATE = BackupCertificate
)
```

- Service Accounts

Use dedicated:

svc_sql_backup --> MSSQL Server svc_prometheus --> Prometheus svc_grafana --> Grafana

- Prometheus Monitoring Design

- Architecture

SQL Server | v MSSQL Exporter | v Grafana Alloy | v Prometheus | v AlertManager | v Grafana

- Metrics

- Backup Age Purpose: Time since last successful backup. Query:

```
sql_backup_last_success_age_seconds
```

Alert: sql_backup_last_success_age_seconds > 86400

- Backup Duration sql_backup_duration_seconds
- Backup Size sql_backup_size_bytes
- Job Failures sql_agent_job_failed_total
- Database Growth mssql_database_size_bytes
- Disk Utilization windows_logical_disk_free_bytes
- Log Growth mssql_log_file_used_percent

- Grafana Dashboard Design

- Executive Dashboard

Audience: Management Widgets: Backup Compliance % Backup Success Rate Failed Backups Capacity Forecast SLA Status

Visualizations: Stat Panels Gauges Pie Charts

- DBA Dashboard

Widgets: Backup Age Backup Duration Backup Size Job Status Restore Tests

Visualizations: Tables Heatmaps Time Series

- Compliance Dashboard
 - Widgets: Retention Compliance Encryption Compliance Restore Validation Status
 - Visualizations: Tables Traffic Lights Compliance Score
- Alerting Design
 - Failed Backup Severity: Critical PromQL : `increase(sql_backup_failed_total[1h]) > 0` Response: Immediate investigation.
 - Missing Backup Severity:Critical PromQL :`sql_backup_last_success_age_seconds > 86400`
 - Backup Duration Anomaly Severity: Warning PromQL : `sql_backup_duration_seconds > (avg_over_time(sql_backup_duration_seconds[30d]) * 2)`
 - Low Disk Space Severity: Critical PromQL : `windows_logical_disk_free_percent < 15`
 - Failed SQL Agent Job Severity: Warning PromQL: `increase(sql_agent_job_failed_total[1h]) > 0`
 - Transaction Log Growth
 - Severity: Warning PromQL: `mssql_log_file_used_percent > 80`
 - Restore Validation Failure Severity: Critical PromQL : `sql_restore_validation_failed_total > 0`
- Disaster Recovery Recommendations
 - Backup Rule
 - Implement:
 - 3-2-1 Rule 3 Copies 2 Different Media Types 1 Offsite Copy
 - Immutable Storage
 - Recommended:
 - Object Lock WORM Storage Immutable Backup Repository
 - DR Testing

Activity	Frequency
Restore Test	Weekly
DR Simulation	Quarterly
Full Recovery Exercise	Yearly
- Implementation Roadmap
 - Quick Wins (0–30 Days) High Priority
 - Deploy Ola Hallengren
 - Enable Compression
 - Implement Encryption
 - Remove xp_cmdshell dependencies
 - Configure backup verification
 - Medium-Term (30–90 Days) Monitoring
 - Deploy Grafana Alloy
 - Deploy MSSQL Exporter
 - Deploy Prometheus
 - Deploy AlertManager
 - Reporting Create Grafana dashboards
 - Long-Term (90+ Days)
 - Enterprise Maturity
 - Immutable storage
 - DR site integration
 - Automated restore lab
 - Capacity forecasting
 - SLA reporting
 - Risk Matrix

Risk	Impact	Likelihood	Priority
Backup Failure	Critical	Medium	High
Disk Exhaustion	Critical	High	High

Encryption Missing	High	High	High
Restore Failure	Critical	Medium	High
Monitoring Gap	High	High	High
Agent Failure	Medium	Medium	Medium
Repository Failure	Critical	Low	Medium
Human Error	High	Medium	Medium

- Conclusion

The current gCABA backup framework is operationally mature and already incorporates several important DBA practices such as Full, Differential, Transaction Log backups, retention management, reporting, and restore validation. However, it relies heavily on custom-developed logic and operating-system-level automation that increases operational complexity, security exposure, and maintenance effort.

The recommended target state is a standardized enterprise backup platform based on SQL Server 2022, Ola Hallengren Maintenance Solution, Prometheus, Grafana Alloy, Prometheus AlertManager, and Grafana. This architecture will provide:

Encrypted backups (AES-256) Automated backup lifecycle management Centralized monitoring and observability Real-time alerting Compliance reporting Improved RPO/RTO achievement Reduced operational risk Simplified maintenance and support

- Add on the corresponding place

Based on the scripts, I already see several important characteristics of the current implementation that should influence the redesign.

- Initial Findings from the Current Solution

- Backup Types Identified

The current methodology includes:

- Full backups
- Differential backups
- Transaction Log backups
- Restore validation jobs
- Backup copy jobs to NFS storage
- Backup purge/retention jobs
- Reporting jobs

- Storage Architecture

The current design appears to use:

Location	Purpose
H:\BACKUP_GCABA	Local backup staging area
K:\BACKUP_GCABA	NFS-mounted backup repository
E:\REPORTES	Reporting and logging

The NFS share is mounted from: **\ARO-NFS\backup\dbbackups**

- Strengths

The current methodology already contains several good practices:

- Separate Full/Differential and TLog backup jobs
- Backup reporting jobs
- Restore testing jobs
- Secondary backup repository (NFS)
- Automated cleanup jobs
- Backup verification logic
- Operational reporting

- Weaknesses

1. Custom Backup Framework The entire solution is heavily based on custom *T-SQL, PowerShell and xp_cmdshell logic*.

- Risks:

- Difficult maintenance
- Knowledge concentrated in few people
- Complex troubleshooting
- Limited standardization

- Recommendation:

Replace with:

- Ola Hallengren Maintenance Solution
- Native SQL Agent schedules
- Centralized configuration tables

2. Extensive xp_cmdshell Usage Found in multiple jobs **exec xp_cmdshell 'dir ...'**.

- Risks:
 - Security exposure
 - Compliance concerns
 - Difficult auditing
- Recommendation:
 - Remove wherever possible.

3. Retention Logic Managed by File Deletion Scripts Current solution uses PowerShell deletion jobs.

- Risks:
 - Accidental deletions
 - No catalog awareness
 - No backup history validation
- Recommendation:
 - Use Ola Hallengren retention parameters: **@CleanupTime**

4. No Backup Encryption Current scripts indicate: **set @cifrado='N'**

5. Monitoring Gap Current monitoring is primarily:

- Text files
 - Email reports
 - SQL Agent Job status
- Missing:
 - Backup age monitoring
 - Backup SLA tracking
 - RPO/RTO visualization
 - Trend analysis
 - Centralized alerting

◦ Recommended Target Architecture

You can view the Grhaptic proposal [\[here\]](#)(./Recommended Target Architecture.drawio)

◦ Backup Strategy recommend

- Full Backups
 - Weekly
 - Sunday 01:00
 - Compression Enabled
 - Encryption Enabled
- Differential Backups
 - Daily
 - Monday-Saturday
 - Every 24 hours
- Transaction Log Backups
 - Every 15 minutes
 - Production databases only
- System Databases
 - Daily: master msdb model

◦ Retention Recommendation

Backup Type	Retention
Log	14 days
Daily	35 days
Weekly	12 weeks
Monthly	13 months

Yearly	7 years
--------	---------

- Monitoring Metrics for Prometheus
 - Critical metrics:
 - Last successful backup age
 - Last successful log backup age
 - Backup duration
 - Backup size
 - Backup throughput
 - Restore test status
 - SQL Agent Job status
 - Disk utilization
 - Database growth
 - Log file growth
 - Failed backup count